List and descriptions of biofilm features calculated by the MicroICS pipeline.

Morphological features:

Feature name in the manuscript	Feature name in the datafile provided by the pipeline	Input	Value normalization	Calculation per _	Description of feature	Threshold
Counts (threshold < n)*	counts(inten<threshold-SUMMARYCustomAlgos.tsv	raw image	no	layer	The number of bacteria in each layer of the 3D image is determined by first applying the threshold value, then segmenting the individual bacteria, and finally counting them. The raw image or an image whose intensity values have been normalized is used as input. For the “normalized” feature, the counts per layer are normalized as a proportion of the sum of all values * 100 (to obtain percentages) per value.	Range from 0.01 to 0.2 increments for 0.01
Counts (normalized intensity, threshold < n)*	counts(Norminten<threshold-SUMMARYCustomAlgos.tsv	intensity normalised on raw image	yes	layer		Range from 0.01 to 0.2 increments for 0.01
Counts (normalized per layer, threshold < n)*	counts(inten<thresholdNormalized-SUMMARYCustomAlgos.tsv	raw image	yes	layer		Range from 0.01 to 0.2 increments for 0.01
Counts (normalized intensity, normalized per layer, threshold < n)*	counts(Norminten<thresholdNormalized-SUMMARYCustomAlgos.tsv	intensity normalised on raw image	yes	layer		Range from 0.01 to 0.2 increments for 0.01
Biovolume	BioVolumeThr-SUMMARYCustomAlgos.tsv	raw image	no	image	The biofilm volume is calculated by counting the number of pixels belonging to the biofilm after thresholding and multiplying this by the voxel size, which represents the physical dimensions of each pixel in height, width, and length (in micrometres). This gives the total volume occupied by the biofilm in the 3D image. Only one volume value is given per image, and the aggregated values remain consistent.	Range from 0.01 to 0.29 increments for 0.01
GPT volume	GPTVolume-SUMMARYCustomAlgos.tsv	raw image	no	layer	The GPT volume is similar to the biovolume described above, but here the biovolume is first determined layer by layer after thresholding, and then the values of these layer counts are aggregated for each image.	Fixed value determined for each image
Normalized GPT volume	GPTVolumeNormalized-SUMMARYCustomAlgos.tsv	raw image	yes	layer		Fixed value determined for each image
Substratum relative coverage	SubstratumRelativeCoverage-SUMMARYCustomAlgos.tsv	raw image	no	image	Percentage of the surface of the well (1st layer of the image) covered with biofilm.	Fixed value determined for each image

**Feature “counts”: Counts are a unique feature that captures variations in counts of bacteria within biofilm by changing the threshold used to detect individual bacteria. It is an ensemble of features known to be correlated with each other and useful in machine learning to capture all aspects of thresholding and counting.*

Statistics-based intensity features:

Feature name in the manuscript	Feature name in the datafile provided by the pipeline	Input type	normalization	Calculation per _	Description of feature	Threshold
Global mean	globalMean-SUMMARYDiffGlobal.tsv	raw image	no	image	The average pixel intensity of the entire 3D image reflects the overall brightness of the biofilm. A single value is calculated for each image, whereby all aggregated values are identical.	no
Layer-wise global maximum**	maxdiffs-SUMMARYDiffGlobal.tsv	raw image	no	image	The average pixel intensities are calculated for each layer of the 3D image, and the difference between the average of each layer and the next is calculated. Only the maximum difference is given, which represents the largest change in brightness between the layers and extends the understanding of the global mean.	no
Layer-wise global minimum**	mindiffs-SUMMARYDiffGlobal.tsv	raw image	no	image	The difference between the average intensities of consecutive layers is calculated in a similar way, but only the minimum difference is given to highlight areas of little or no change, which may correspond to stable biofilm regions or the background.	no
Layer-wise global mean**	mdiffs-SUMMARYDiffGlobal.tsv	raw image	no	image	The mean difference between the intensities of successive layers is calculated and reported and reflects the overall uniformity or variability of brightness in the vertical dimension of the biofilm.	no
/	eigen-SUMMARYDiffGlobal.tsv	raw image	no	/	A value representing the differences in the overall 3D image is noted but is not currently calculated.	no
Intensity differences**	diff-SUMMARYCustomAlgos.tsv	raw image	no	layer	The difference between maximum and minimum pixel intensity within each layer is calculated and aggregated, where: (a) the mean aggregation indicates the image contrast, b) the median reflects the typical intensity, c) the standard deviation indicates the variability of the contrast between the layers, and d) the quantiles describe the distribution skewness of the contrast between the layers.	no
Normalized intensity differences**	diffNormalized-SUMMARYCustomAlgos.tsv	raw image	yes	layer	The differences between the maximum and minimum intensities per layer are normalized as a percentage of the total max-min sum across all layers and then aggregated to characterize the relative contrast distribution.	no
Mean intensity	mean-SUMMARYCustomAlgos.tsv	raw image	no	layer	The mean pixel intensity is calculated for each layer and then averaged across the layers, representing the average brightness per layer; higher values indicate denser or brighter regions and vice versa.	no
Normalized mean intensity	meanNormalized-SUMMARYCustomAlgos.tsv	raw image	yes	layer	The mean intensity on each layer is normalized as a percentage of the total sum of mean intensities per image before averaging across layers.	no
Median intensity	med-SUMMARYCustomAlgos.tsv	raw image	no	layer	The median pixel intensity is calculated for each layer, and then the median values are aggregated across all layers.	no

Feature name in the manuscript	Feature name in the datafile provided by the pipeline	Input type	normalization	Calculation per _	Description of feature	Threshold
Normalized median intensity	medNormalized-SUMMARYCustomAlgos.tsv	raw image	yes	layer	The median intensity per layer is normalized as a percentage of the total sum of the median intensities per image and then aggregated across layers.	no
Maximum intensity	max-SUMMARYCustomAlgos.tsv	raw image	no	layer	The maximum pixel intensity is calculated per layer and aggregated, highlighting localized bright spots such as clumps or active cells.	no
Normalized maximum intensity	maxNormalized-SUMMARYCustomAlgos.tsv	raw image	yes	layer	The maximum intensity per layer is normalized as a percentage of the total maximum per image and then aggregated.	no
Minimum intensity	min-SUMMARYCustomAlgos.tsv	raw image	no	layer	The minimum pixel intensity per layer is calculated and aggregated to capture the darkest or least dense biofilm regions; in combination with other statistics, it shows biofilm gaps, heterogeneity and spatial structure.	no
Normalized minimum intensity	minNormalized-SUMMARYCustomAlgos.tsv	raw image	yes	layer	The minimum intensity per layer is normalized as a percentage of the total minimum sum per image before aggregation.	no
Standard deviation of intensity	std-SUMMARYCustomAlgos.tsv	raw image	no	layer	The standard deviation of pixel intensities per layer is calculated and aggregated; low values indicate uniform intensity (low contrast), while high values indicate high contrast and high variability.	no
Normalized standard deviation of intensity	stdNormalized-SUMMARYCustomAlgos.tsv	raw image	yes	layer	The standard deviation per layer is normalized as a percentage of the total standard deviation sum per image and then aggregated.	no

***Layer wise features:*
Layer wise features capture how much the intensity fluctuates between the layers and how features differ across the volume.

Texture features:

Feature name in the manuscript	Feature name in the datafile provided by the pipeline	Input type	normalization	Calculation per _	Description of feature	Threshold
Void fraction (min Prop)	minProp-SUMMARYCustomAlgos.tsv	raw image	no	layer	The number of pixels with intensity below the average pixel intensity is counted for each image. Min prop refers to dark areas, such as empty regions not occupied by biofilm.	no
Normalized void fraction	minPropNormalized-SUMMARYCustomAlgos.tsv	raw image	yes	layer		no
Homogeneity	Homogenity-SUMMARYCustomAlgos.tsv	raw image	no	image	Homogeneity is calculated using the co-occurrence matrix, which captures the distribution of intensity changes between neighbouring pixels in the X, Y and Z directions (after Beyenal et al. 2004, doi: 10.1016/j.mimet.2004.08.003). It measures how similar neighbouring image objects are, with higher values indicating greater uniformity. One value is given per image, and the aggregated values are identical.	no
Thickness	ThicknessThreshold=0.01-SUMMARYCustomAlgos.tsv	raw image	no	image	Pixel intensities are tracked vertically from the first image layer, recording for each pixel the height at which the intensity drops to zero. The average of these heights is reported as a single image value using the same aggregation.	Range from 0.01 to 0.29 increments for 0.01

Feature name in the manuscript	Feature name in the datafile provided by the pipeline	Input type	normalization	Calculation per _	Description of feature	Threshold
Roughnes	RoughnessThreshold=threshold-SUMMARYCustomAlgos.tsv	raw image	no	image	The average height difference between neighbouring pixels at a given layer thickness is calculated and reflects the roughness of the biofilm surface. One value per image is given, with uniform aggregation.	Range from 0.01 to 0.29 increments for 0.01
Horizontal spreading	SpreadingHorizontal-SUMMARYCustomAlgos.tsv	raw image	no	image	Three distributions (Dx, Dy, Dz) of the x, y, and z coordinates of the biofilm pixels are created, as described by Mueller et al. (2006, doi: 10.1186/1472-6785-6-1), and the variances for each axis are calculated. Horizontal spreading is calculated by summing up variances in x and y. Vertical spreading (in z direction) equals the variance of z. Total spreading (in z direction) is calculated by summing up variances in x, y and z.	no
Total spreading	SpreadingTotal-SUMMARYCustomAlgos.tsv	raw image	no	image		no
Vertical spreading	SpreadingVertical-SUMMARYCustomAlgos.tsv	raw image	no	image		no
GPT fractal dimension	GPTFractalDim-SUMMARYCustomAlgos.tsv	raw image	no	layer	The fractal dimension reflects the complexity and irregularity of the surface or boundary of the biofilm and quantifies this complexity at different scales; higher values indicate more complex and branched structures rather than smooth shapes.	no
Normalized GPT fractal dimension	GPTFractalDimNormalized-SUMMARYCustomAlgos.tsv	raw image	yes	layer		no
GPT homogeneity	GPTHomogeneity-SUMMARYCustomAlgos.tsv	raw image	no	image	This feature is calculated similarly to non-GPT homogeneity, but is first computed for each layer and then aggregated for each image; higher homogeneity indicates a more uniform and consistent texture.	no

The following features were calculated by Graycomatrix method, which constructs a grayscale co-occurrence matrix (GLCM) by calculating how many times a pixel with intensity value (grayscale) *i* occurs in a given spatial relationship to a pixel with value *j*.

Feature name in the manuscript	Feature name in the datafile provided by the pipeline	Input type	normalization	Calculation per _	Description of feature	Threshold
GPT contrast	GPTContrast-SUMMARYCustomAlgos.tsv	raw image	no	image	Contrast emphasises the difference between grey-level values. Larger differences in intensity contribute more to the sum, so a higher contrast value indicates greater variability in the intensities of neighbouring pixels.	no
GPT correlation	GPTCorrelation-SUMMARYCustomAlgos.tsv	raw image	no	image	The correlation value is calculated similarly to the Pearson correlation coefficient, using the correlations between values in the graycomatrix. High correlations indicate that certain grey-level pairs occur in a predictable manner.	no
GPT dissimilarity	GPTDissimilarity-SUMMARYCustomAlgos.tsv	raw image	no	image	Similar to contrast, but it is calculated from absolute differences instead of a squared difference. Dissimilarity also captures local variations in gray levels, but its contribution grows more linearly compared to contrast’s squared term. Dissimilarity is opposite to homogeneity.	no
GPT energy	GPTEnergy-SUMMARYCustomAlgos.tsv	raw image	no	image	Energy is another way to express the uniformity or orderliness in the texture. It is often used because it is on a [0,1] scale if P is normalized.	no